

OPERATIONS HEALTH ASSESSMENT REPORT - TIKINOW

TikiNow Platform · Dataset: 629,064 orders · Year: 2017

1. EXECUTIVE SUMMARY & KEY OPERATION KPIs

The operational data for TikiNow in 2017 reveals a critical paradox: a high Completion Rate of 79.98% masks severe systemic failures in service level commitments (SLA), where nearly half of all orders missed their delivery deadlines and roughly 1 in 5 orders were cancelled before completion.



- **On-Time Delivery (OTD) Rate (52.33%):** This indicates that nearly half of all orders violated the promised Service Level Agreement (SLA). This is the most critical metric requiring immediate operational intervention.
- **Cancellation Rate (19.74%):** Representing 1 in 5 orders, this high cancellation volume signals significant revenue leakage and potential customer dissatisfaction.
- **Failed Delivery Rate (3.49%):** This stands out as the most stable operational metric, remaining within acceptable industry thresholds.

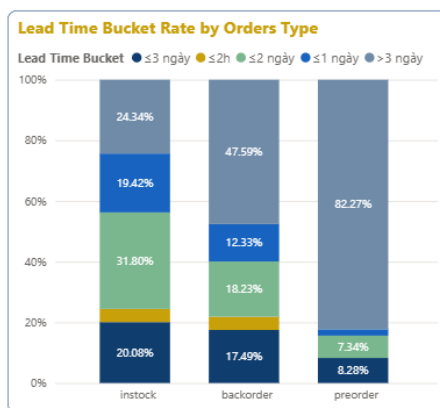
2. DELIVERY PERFORMANCE & LEAD-TIME ANALYSIS

2.1. TikiNow Express (2h) — Critical SLA Breach

#	Order Type	On Time Rate	STATUS
1	Non-Express	53.46%	Moderate
2	Express (2h)	9.58%	CRITICAL

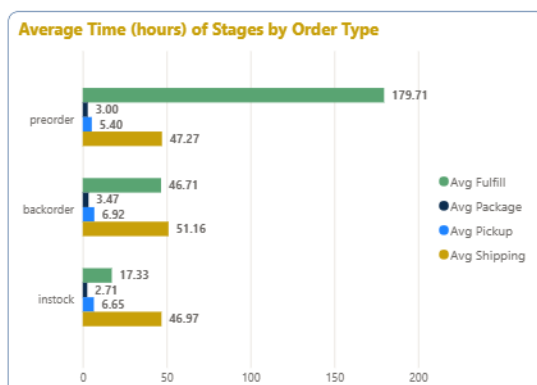
➡ **Key Insight:** The On-Time Delivery rate for the Express (2h) service is a catastrophic **9.58%**. This means 9 out of 10 Express orders failed to meet the 2-hour commitment, directly threatening brand equity and the core value proposition of TikiNow.

2.2. Lead-Time Distribution by Order Type



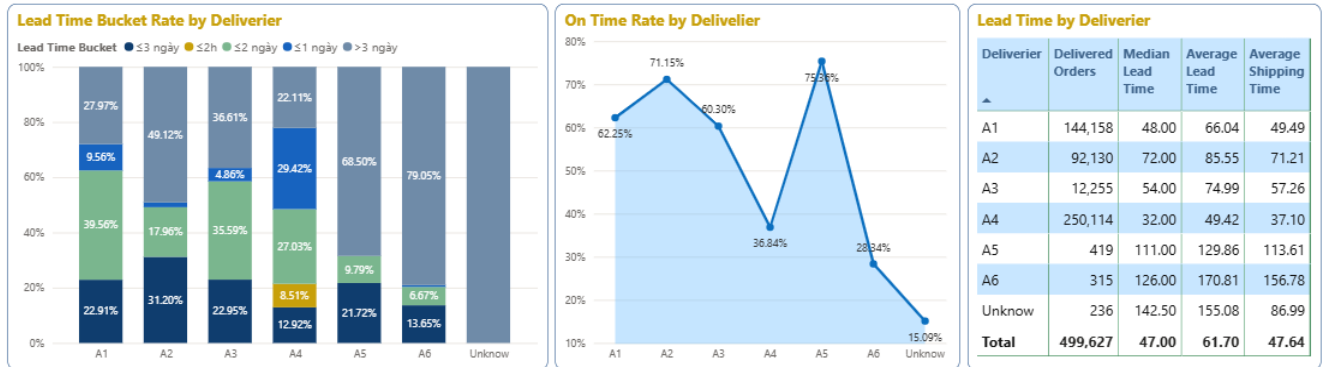
- **Instock:** Demonstrated the highest processing efficiency, with a cumulative **23.76%** of completed orders processed within ≤ 1 day. The total cumulative percentage of orders delivered within ≤3 days reaches an impressive 75%.
- **Backorder:** Performance decreased to only 12.33% of orders completed within 1 day, and the total percentage delivered within 3 days reached 52.41%.
- **Pre-order:** A sharp decrease of only 17.73% of total orders completed within the timeframe of less than 3 days, reflecting the specific characteristics of product groups that require waiting time from suppliers.

2.3. Logistics Lifecycle Bottleneck: In-Warehouse vs. Last-Mile



- **Data Insight:** Across all order types (Instock, Backorder, Preorder), the actual time spent on the road (Shipping Time) constitutes **85% (Median)** and **77.21% (Average)** of the total fulfillment Lead-Time.
- **Operational Conclusion:** Internal warehouse fulfillment (pick, pack, dispatch) is relatively well-optimized. The definitive bottleneck lies entirely within **Last-Mile Delivery**. Furthermore, the Last-Mile shipping time for Instock orders (~47 hours) is nearly identical to that of Backorders (~51 hours), indicating a lack of priority routing for readily available items.

3. LAST-MILE CARRIER PERFORMANCE EVALUATION



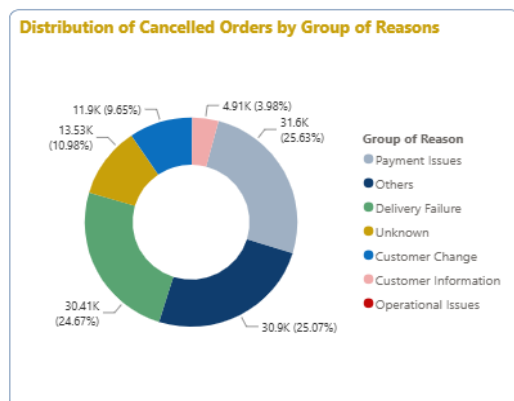
Note: Delivered Orders by Carrier in this table (499,627 orders) are lower than the total number of successfully delivered orders by status Completed (503,118 orders) because orders with Lead Time ≤0 have been excluded to ensure the accuracy of the average indicators.

Strategic Carrier Performance Segmentation:

- **High-Performing Partners (A1, A2 & A3):** Carrier A2 (71.15% OTD) maintains a competitive on-time rates and A1 & A3 (> 60% OTD) are the most reliable partners, maintaining short shipping times.
- **The Primary Operational Critical priority (A4):** Carrier A4 absorbs the highest volume (~50% share with 250k orders with 1/5 is a 2h order) but delivers a dismal OTD rate of 36.84% (This stems from A4 handling 2h express deliveries with extremely low on-time rate.). A4 is single-handedly pulling down the entire platform's SLA.
- **Long-Tail Noise (A5 & A6):** Carrier A6 has the worst performance (28.34% OTD & 156.78 hours average shipping), but represents only 315 orders – 0.06% of total delivered. While operationally deficient, its minimal volume means it is a low-priority issue compared to A4.

4. ROOT CAUSE ANALYSIS (RCA) FOR ORDER FAILURE

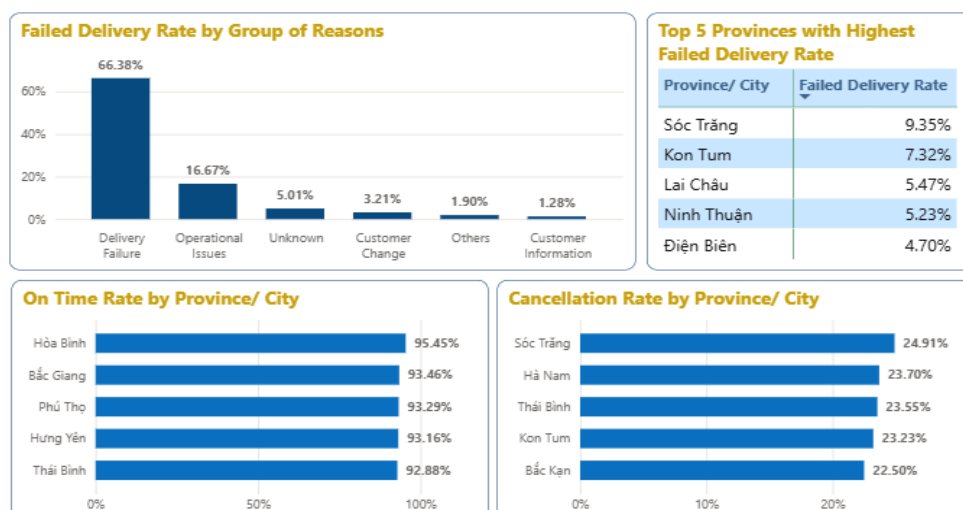
4.1. Order Cancellation Drivers (19.74% of Total Orders)



The reasons for order cancellations are concentrated in three main factors, accounting for approximately 75% of all cancellations:

- **Payment Issues:** (25.63%): a mixed-cause category encompassing both platform-side failures (payment gateway errors, transaction timeouts) and customer-side drop-offs (insufficient funds, card declines).
- **Delivery Failure:** 30,412 orders (24.67%).
- **Systemic Data Gaps:** A combined 36.05% of cancellations fall under "Others" (25.07%) and "Unknown" (10.98%), exposing a significant gap in order-tracking governance.

4.2. Failed Delivery Drivers (3.49% of Total Orders)



- **The Overwhelming Factor:** 66.38% of failed deliveries are logged under a generic "Delivery Failure" reason code.
- **Operational Misalignment:** "Operational Issues" account for 16.67% of failures, highlighting internal friction during handover to carriers.
- **Top 5 provinces** with the highest on-time performance (93% or higher): These provinces are all located in the northern region, demonstrating superior optimization by warehouses or shipping partners in this regional cluster.
- **Soc Trang & Kon Tum:** Emerged as high-risk regions suffering from a dual crisis: exceptionally high Failed Delivery Rates (Sóc Trăng: 9.35%, Kon Tum: 7.32%) compounded by severe Cancellation Rates (~23% - 25%). Last-mile networks in these regions require localized auditing.

5. STRATEGIC RECOVERY & ACTION PLAN

5.1. PRIORITY MATRIX

#	Subject	Measurable Metric (KPI)	Priority	Object
1	Express (2h): Massive delays	On-time: 9.58%	 CRITICAL	Ops + Logistics
2	Carrier A4: Largest, worst	A4: On time 36.84% / 250k orders	 CRITICAL	3PL Partnership
3	Last-mile accounts for 85% (median) and 77.21% (Avg) of total lead time	Median_shipping: 40.00h Avg_shipping: 47.64h	 HIGH	Last-mile Ops
4	Payment Issues = 25.63% of orders cancellation	Cancel rate: 19.74%	 HIGH	Product/Tech
5	Cancellations with unknown reasons	Unknown (Null) + Others: 36.05%	 HIGH	Data/Ops
6	Soc Trang: Double (fail + cancel)	Fail: 9.35% / Cancel: 24.91%	 MEDIUM	Regional Logistics
7	Carrier A6: Excessive lead time	On time 28.34% / Avg_ship 156.78h	 MEDIUM	3PL Partnership

5.2. ACTION PLAN

1	CRITICAL - Express Emergency Response (2 hours) & Carrier A4 Audit and Improvement • Implement emergency RCA (Root Cause Analysis) for warehouse → shipper handover process for Express orders. Temporarily narrow Express coverage to central HCM/HN areas until capacity is sufficient. Set interim KPIs: on-time Express ≥40% in Q1, ≥70% in Q2. • A4 is currently the biggest systemic risk: Review contracts, set a minimum SLA of 55% on-time with clear penalties. Gradually distribute 15-20% of A4 volume to groups with A2 and A3 performance to reduce dependence on partner orders.
2	HIGH - Improved Payment Funnel & Reduced Cancellations • 25.63% cancellations from Payment Issues: This represents the single largest addressable cancellation driver. Implement automated payment retry flow, add payment reminders (push notification + SMS), added E-wallets/ COD for high-cancellation groups. • Add mandatory reason code when shippers/ CSRs process cancellations to minimize the 36% Unknown (Null)/Others gap. • Separate last-mile processing lanes, shippers prioritize consolidation and delivery of Instock orders and especially Express packages (2 hours) to shorten Average Shipping.
3	MEDIUM - Addressing Geographic Gaps & Lead Time for A6 • Soc Trang & Kon Tum face a dual problem: high failed deliveries and high cancellations. Re-evaluate local carriers in these provinces; consider specialized delivery partners for remote areas. Experiment with a pickup-point delivery model in areas with complex terrain. • Review contracts with A6; focus on addressing bottlenecks in the transit phase to reduce Average Shipping Time.

6. DATA QUALITY WARNING & SYSTEMIC ANOMALIES

- To preserve data integrity, systemic data anomalies were selectively isolated rather than completely deleted:

Identified Anomaly	Impacted Volume	Metric Affected	Operational Mitigation Applied
purchase_at > delivery_deadline	52,819	On Time Rate	Excluded from OTD calculations; flagged as a critical system timestamp logic error.
purchase_at > cancelled_at	35,380	Cancellation Lead-Time	Retained in dataset & flagged as a warning
purchase_at > actual_delivery	138	Carrier Performance / Lead Time	Filtered out (lead_time > 0) specifically in Section 3 to prevent skewing partner SLA metrics.
Test data inside "reason_cancel/fail_delivery"	468	Root Cause Analysis	Excluded from percentage distribution charts.
actual_delivery: year listed as 2006/2007	86	Lead Time Timeline	Dynamically synchronized to the baseline year (2017)
Status is "Completed" but contains cancelled_at	11	Completion Rate/ On Time Rate	Excluded from OTD & Completion Rate calculations
Status is "Completed" but actual_delivery is Null	4,484	Lead Time/ On Time Rate	Filtered out Lead_time > 0
Status is "Cancelled" but cancelled_at is Null & Order_value < 0	11,365/ 5		Retained in dataset

- Final DA Note to IT/ Engineering:** The 52,819 rows with inverse SLA timestamps represent **~8.4%** of the entire dataset. This exceeds acceptable data-error thresholds (>5%) and demands an immediate engineering review of the platform's order timestamp logging logic.
- The streams that affect the analysis are selectively processed on a metric-by-metric basis - not entirely removed because the cause is not yet clearly identified and to preserve the integrity of the analysis.

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